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AIM 2025 Challenge on Screen-Content Video Quality Assessment: Methods and Results

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Abstract

This paper presents an overview of the AIM 2025 Challenge on Screen Content Video Quality Assessment. The challenge included a set of 150 source videos. To receive distorted versions, the source videos were transmitted through video conferencing applications, introducing real-world distortions such as compression artifacts and frame drops. Distorted versions were labeled by human crowdsourcing assessors to receive reference subjective scores. The evaluation was based on subjective quality assessment via crowdsourcing, obtaining votes from over 8,000 assessors. The goal of the participants was to develop an algorithm to assess the visual quality of the videos, achieving the highest correlation with the subjective scores. The challenge attracted more than 45 registered teams, 5 of which passed the final phase with source code verification. The outcomes may provide insights into the state of the art in screen-content video quality assessment and highlight emerging trends and effective strategies in this evolving research area. All data, including the processed videos and subjective comparison votes and scores, is made publicly available — https://github.com/msu-video-group/AIM25_SC_Quality_Assessment

1. Introduction

Screen content video, which includes slide presentations, document displays, code editors, and graphical user interfaces, has become an integral component of contemporary digital communication. Its prevalence is particularly notable in contexts such as remote work, online education, and collaborative software development, where screen sharing is commonly employed. In contrast to natural video content, screen-content typically contains static regions, sharp edges, and high-density textual information. These distinct visual properties present unique challenges for video quality assessment.

To approximate human perception of video quality, objective video quality assessment (VQA) techniques rely heavily on subjective ground truth. Conventional full-reference VQA models such as PSNR, SSIM, and VMAF depend on having perfectly aligned original and degraded video sequences. Yet in screen-conten scenarios, especially those involving real-time transmission, frame misalignment is common due to frame drops or network-induced irregularities. This misalignment undermines the reliability of full-reference methods, requiring additional preprocessing for temporal synchronization, and often still yielding limited accuracy.

Recent advancements have led to a surge in no-reference VQA algorithms, many of which are deep learning-based [16, 18, 28, 30–32]. However, these models are typically optimized for natural video and lack evaluation on screen content. The temporal and spatial dynamics of screen-content videos are markedly different, resulting in poor generalization and reduced accuracy when such models are applied outside their original domain [20].

This challenge aims to provide a platform for researchers and industry professionals to develop and evaluate algorithms for screen-content video quality assessment. The objectives of this Screen Quality Video Quality Assessment are to establish a benchmark for screen-content video quality assessment on real-world videos transmitted through videoconferencing applications and containing diverse distortions. Also the goal is to encourage the development of algorithms that could assess perceptual quality of screen-content videos with high reliability, helping screen-content transmission optimization.

In this competition, we used a dataset of 150 source videos to evaluate the submitted methods. There were 2400 distorted videos with subjective scores. For each dis-

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torted video, a mapping between the distorted frame indices and the corresponding original frame indices was provided. This mapping is necessary because video transmission through conferencing applications, particularly under constrained network conditions, often results in dropped or delayed frames. The dataset includes diverse content types and videos were transmitted under different network conditions, reflecting the real-world scenarios of screen-content video viewing.

Challenge dataset was split into training, validation, and testing sets with sizes of 70, 30, 50 videos for training, validation and test stages, respectively. From the test set, 30 videos were available to the participants, the remaining 20 were in a private subset and were released only after the end of the competition. Methods results on these 20 videos were obtained by the organizers team independently by running the code of the participants solutions in the final phase of the competition. The final evaluation dataset consists of 480 distorted videos with corresponding subjective scores.

To evaluate participants methods at all phases of the competition, we used Spearman, Pearson and Kendall correlations with the subjective scores. Subjective scores were aggregated using the Bradley-Terry model [3]. This approach ensures a robust subjective assessment of methods by leveraging both direct quality comparisons and ranking-based score estimation.

The competition consists of the development and final test stages, attracting a total of 45 registered participants. Across these stages, 9 teams participated. Ultimately, 5 teams provided fact sheets and passed source code verification in the final stage. Descriptions of the methods proposed by the participants are provided in Section 4.

This challenge is one of the AIM 2025 [1] workshop associated challenges on: high FPS non-uniform motion deblurring [4], rip current segmentation [8], inverse tone mapping [27], robust offline video super-resolution [15], screen-content video quality assessment [21], real-world raw denoising [5], perceptual image super-resolution [6], efficient real-world deblurring [10], 4K super-resolution on mobile NPUs [11], efficient denoising on smartphone GPUs [13], efficient learned ISP on mobile GPUs [12], and stable diffusion for on-device inference [14]. Descriptions of the datasets, methods, and results can be found in the corresponding challenge reports.

2. Challenge

The AIM 2025 Screen Content Video Quality Assessment Challenge is organized to advance research on evaluating the perceptual quality of screen content videos. This challenge focuses on developing and benchmarking methods that can accurately predict human-perceived quality for videos containing text, graphics, and mixed screen content. By providing a large-scale dataset and standardized evaluation protocols, the challenge establishes a reliable benchmark to guide future progress in screen content video quality assessment. The following sections describe the dataset, evaluation methodology, and competition phases in detail.

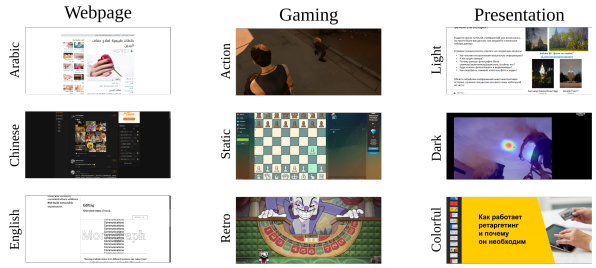


Figure 1. Dataset video examples.

2.1. Dataset

For the evaluation, we employed the dataset publicly released in [20]. Following the methodology described in that work, we additionally generated 50 videos to extend the testing set. This section outlines the methodology used to develop the dataset. To simulate transmission and compression artifacts, each source video was processed using 16 different distortion settings, yielding a total of 2400 degraded sequences. The distortion processes include screen recordings of video conferencing platforms operating under constrained network conditions, as well as standard video compression codecs.

To enable rigorous quality evaluation, we acquired over 60,000 pairwise quality judgments from a pool of more than 8,000 unique human participants. These comparisons were analyzed using the Bradley–Terry model [2] to derive a consistent quality ranking across the dataset. Frame-level alignment was performed between each distorted sequence and its original counterpart. The dataset is created for evaluating both reference-based and blind VQA methods under realistic screen content transmission scenarios.

To ensure content diversity and real-world relevance, we captured a wide range of screen-based activities reflecting common use cases. The source videos include sequences of slide navigation, terminal interaction, code editing, web browsing, spreadsheet manipulation, data visualization, application interfaces, and light gaming. Variations in text size, background layout, color schemes, and temporal dynamics such as cursor movement and typing were deliberately introduced to improve representativeness. Figure 1 shows dataset content examples.

All sequences were recorded in FullHD resolution at 30 frames per second using OBS Studio. Recordings were stored using lossless encoding and 4:4:4 chroma sub-

[1] https://www.cvlai.net/aim/2025/

sampling to preserve original visual fidelity and color integrity. Distorted videos were produced through two primary methodologies: screen recording of streaming sessions over video conferencing platforms, and lossy encoding with standard video codecs.

To capture streaming artifacts, we employed OBS Studio to record screen content transmitted through popular conferencing applications. All recordings were conducted using lossless settings and 4:4:4 chroma subsampling. Color configurations were matched to the source to ensure bit-exact reconstruction in ideal cases. Some recordings were cropped and then rescaled to FullHD resolution, leading to spatial misalignment with the original content.

Controlled network degradation was introduced using the `tc-netem` [17] utility, which enabled emulation of packet loss and bandwidth restrictions on the transmitting system.

Subjective assessments were conducted using pairwise comparison methodology, implemented via a crowdsourcing platform. In each task, participants were shown two videos side by side and asked to select the one with superior visual quality. An additional option, "cannot decide," was also provided. Each participant completed 20 such comparisons, which included two control pairs with known quality differences generated by applying high-compression settings to the original videos.

Participants were not informed of the nature of the videos or the presence of control items. Only responses from users who correctly answered both control comparisons were retained for analysis. To maintain statistical balance, each video pair received exactly ten valid votes. In total, responses were collected from more than 8,000 distinct participants, forming a comprehensive foundation for deriving reliable quality rankings.

2.2. Evaluation

In this challenge, we adopted a standard evaluation protocol to assess each method from two complementary perspectives: prediction monotonicity and prediction accuracy. Prediction monotonicity was quantified using the Spearman rank-order correlation coefficient (SROCC) and the Kendall rank-order correlation coefficient (KROCC), which measure how well the predicted scores preserve the ranking of the ground-truth subjective scores. Prediction accuracy was evaluated using the Pearson linear correlation coefficient (PLCC), which captures the strength of the linear relationship between predicted and ground-truth scores. The primary evaluation metric was the Spearman rank-order correlation coefficient (SROCC).

3. Results

In the first development phase of the challenge, 9 participants successfully submitted their results. The outcomes of the validation phase, including intermediate performance metrics, are publicly available in the challenge repository to ensure transparency and reproducibility. For the final evaluation stage, we received 5 valid submissions that satisfied all requirements of the competition protocol.

Table 1. Results of the **AIM 2025 Screen Content Video Quality Assessment Challenge**. Spearman, Pearson and Kendall correlations are computed on the 30 test sequences. Final rankings are determined by Spearman rank-order correlation coefficient (SROCC).

Rank	Team	SROCC ↑	PLCC ↑	KROCC ↑
1	Dreamer	0.9045	0.9574	0.7928
2	lfovains	0.8471	0.9121	0.7133
3	pravin_pardhi	0.8402	0.9205	0.6956
4	ob1to	0.8198	0.8890	0.6832
5	410lab	0.7648	0.8798	0.6417

A detailed overview of the methodologies adopted by the participating teams is provided in Section 4, while comprehensive information on team members and affiliations can be found in Section 5.

Table 1 summarizes the final results obtained on the testing set. The ranking is based on the overall evaluation of 30 test videos, reflecting both consistency across content and robustness of the proposed approaches. This challenge provides a clear comparison of the solutions and establishes a solid foundation for future research in screen content video quality assessment.

4. Teams and Methods

4.1. Dreamer Team

Our team proposes a no-reference multi-modal feature fusion model for screen content video quality assessment. This model integrates textual, visual, and temporal features to accurately capture the quality perception factors of screen content videos without relying on reference videos.

Before training, we process key frames using a frame content selector to identify representative key frames by calculating the structural similarity (SSIM) between consecutive frames. If the similarity falls below a predefined threshold, the frame is selected. Each video has 8 to 12 key frames; if fewer than 8 frames are selected, duplicates from the end are appended to meet the minimum requirement [1, 19].

As shown in the Figure. 2, the model comprises branches for textual and visual semantic analysis, as well as a Temporal Analysis Branch. The textual and visual semantic branches both take the key frames as input. The textual branch uses the EasyOCR framework to extract text regions and content characteristics, thus assessing text readability and display quality [26]. This branch is particularly impor-

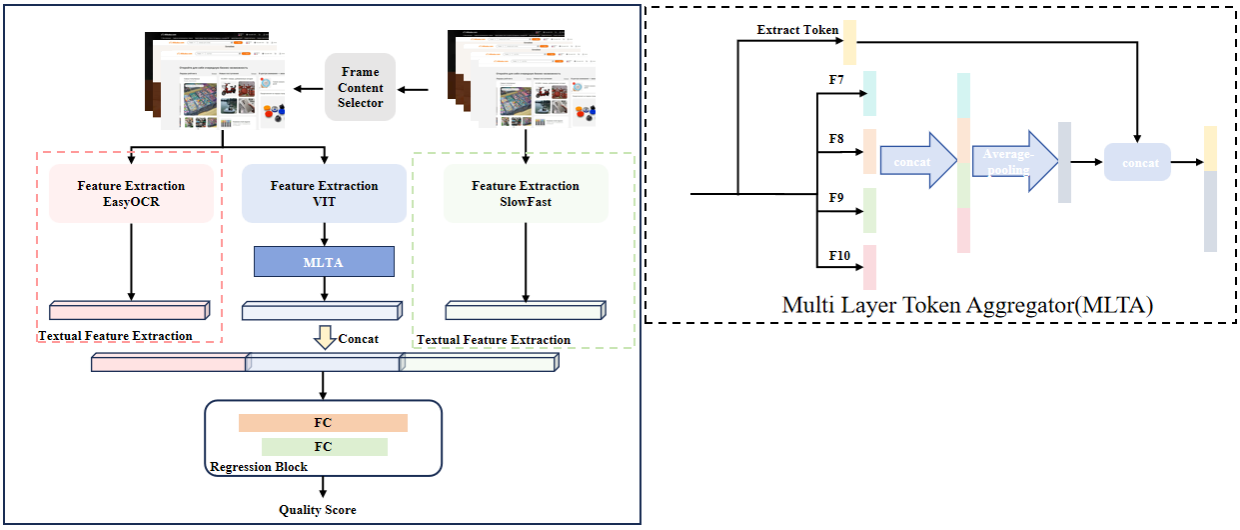


Figure 2. Dreamer team overall pipeline

tant as it detects and evaluates the text present in the screen content, which is a crucial factor in determining the overall quality of the video. By analyzing the clarity and readability of the text, the model can better assess the quality of the video, especially in scenarios where text is a significant component of the content.

The visual semantic branch utilizes the Vision Transformer (ViT-B/16) to extract semantic visual features from the key frames [7, 29]. To enhance multi-layer semantic modeling, a Multi-Layer Token Aggregator (MLTA) is introduced, which aggregates Patch Token features from the 7th to 10th layers of ViT, along with the final Class Token, and fuses them via concatenation and average pooling, thereby enriching the robustness and expressiveness of semantic features. The Temporal Analysis Branch, based on the SlowFast network, captures local and global dynamic variations in screen content, modeling motion patterns such as sliding, switching, and interactive actions over time [24]. The outputs from all branches are concatenated into a unified multi-modal feature vector, which is then fed into a regression block composed of two fully connected layers to predict the final video quality score.

For training, key frames are cropped to a resolution of 384×384 pixels and fed into the branches for textual and visual semantic analysis. The full video is processed by the Temporal Analysis Branch. We employ data augmentation strategies, including random cropping and normalization, to enhance the robustness of the model. To further improve training efficiency, we introduce a linear learning rate decay schedule. The model is trained on a single NVIDIA Tesla V100S GPU with a batch size of 3 for 30 epochs. We use the Adam optimizer with an initial learning rate of 1×10^{-5}, which decays by a factor of 0.9 every 10 epochs. Our model achieves strong performance on the validation set, as measured by the SRCC, PLCC, and RMSE metrics. We train our model on the training set provided by the competition and select the optimal model checkpoint based on the performance metrics on the validation set. During the testing phase, we employ the same data pre-processing and input sampling strategies as in the training phase to ensure consistent input distributions and feature representation capabilities across different stages.

4.2. ifovains Team

We propose a hybrid no-reference Video Quality Assessment (VQA) pipeline method to estimate video quality. Our model integrates hand-crafted and deep learning-based features to predict the visual quality of screen content videos.

We extract the frames from the video after converting dataset videos from MP4 format to raw YUV format.

4.2.1. Swin-Tiny Spatial Modeling

Frames undergo custom preprocessing (edge enhancement, contrast adjustment, and optional deblurring) before being passed through a fine-tuned Swin-Tiny Transformer. Patch-level embeddings are extracted from the penultimate transformer block and spatially averaged.The Swin-T encoder produces a 768-dimensional vector feature per video after frame-wise pooling.

4.2.2. Vision Mamba + LSTM Temporal Modeling

A pre-trained ViM-Tiny model [9] extracts CLS token embeddings per frame. These are passed through a two-layer LSTM to model temporal dependencies. The final

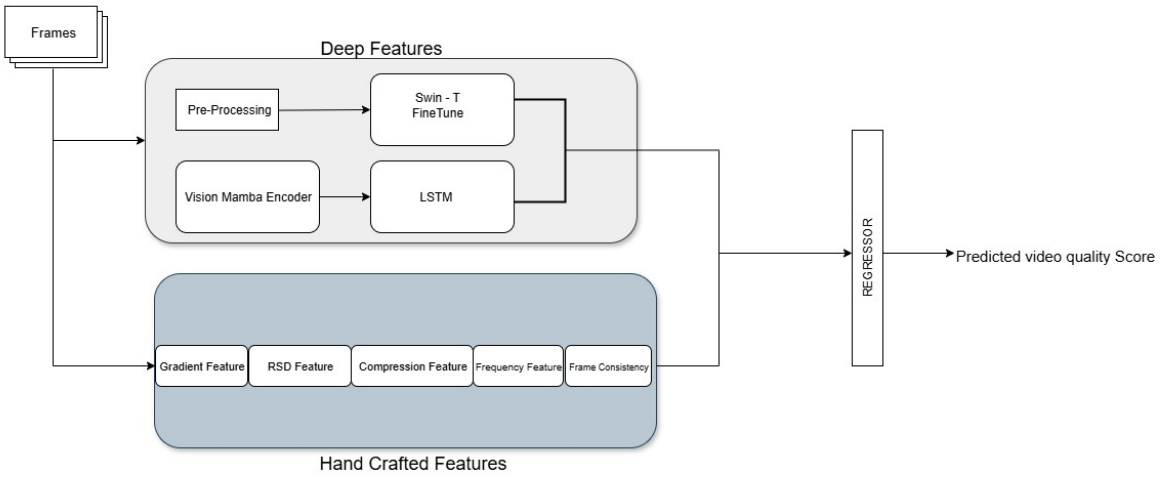


Figure 3. ifovains Team solution: Architecture of the proposed VQA pipeline combining Swin-T, Vision Mamba + LSTM, and handcrafted features

representation is enriched with statistical pooling (mean, std, min, max, and segmental means), resulting in a 1792-dimensional temporal feature vector of per video.

These two deep features are then concatenated to give 2560-dimensional feature vector per video.

4.2.3. Handcrafted Feature Extraction

We extract interpretable, frame-level grayscale features and aggregate them across time [25]:

- **Gradients:** Std and entropy of Sobel-based edge maps.
- **Structural Distortion (RSD):** Residuals between the frame and its Gaussian-blurred version.
- **Compression Artifacts:** Block-edge sensitivity and corner statistics.
- **Wavelet Energy:** Frequency decomposition.
- **Temporal Change:** Frame difference energy to capture flicker or scene cuts.

All frame-level features are flattened to a single video-level feature. We took the maximum feature vector and zero padded where needed to ensure the dimensional consistency.

4.2.4. Feature Fusion and Regression

The combined vector is standardized and input into ensemble gradient boosting regressors (XGBoost and LightGBM), each trained to predict the Mean Opinion Score (MOS). The final predicted MOS is obtained by averaging the outputs of both models.

4.3. pravin_pardhi Team

The proposed method uses a hybrid feature fusion strategy that combines:

- Deep features from **ResNet-18** pretrained on ImageNet and frozen for stability.
- **Handcrafted features** capturing distortion-specific statistics like sharpness and gradient magnitudes from the middle frame of the video.
- A fully connected **MLP regressor** to predict subjective quality scores.

The model is trained with a hybrid loss combining **MSE loss and Spearman rank correlation loss** to balance accuracy and monotonic ranking. During inference, only the middle frame is used, making the approach efficient and scalable.

4.3.1. Deep Features

We use ResNet-18 pretrained on ImageNet. The final fully connected layer is removed, and the output from the global average pooling layer is used. The backbone is frozen to avoid overfitting on the SCV dataset, following design motivations similar to other UGC-focused deep learning approaches [23].

4.3.2. Handcrafted Features

We extract 5 frame-level features from the grayscale middle frame:

- Laplacian variance (sharpness)
- Gradient mean
- Gradient standard deviation
- Maximum gradient
- Minimum gradient

These features help model edge contrast, screen sharpness, and subtle SC distortions often missed by generic deep models [34].

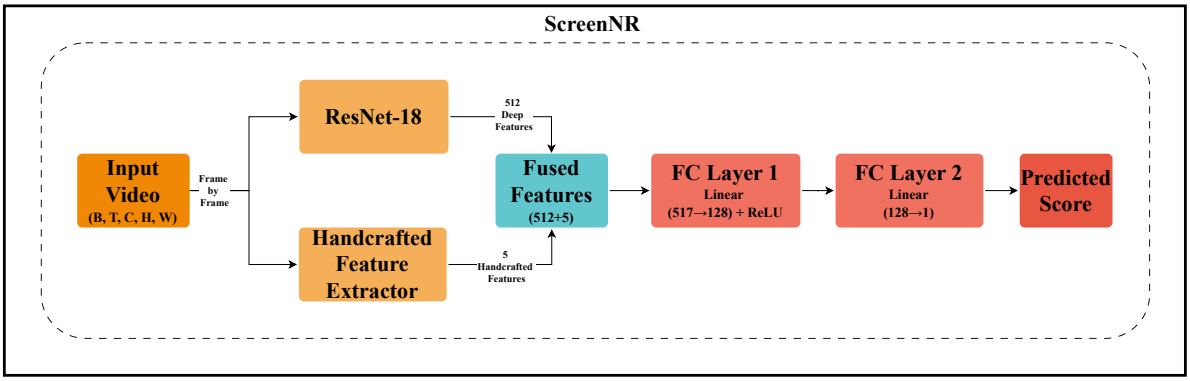


Figure 4. pravin_pardhi Team ScreenNR-VQA Pipeline: Deep and handcrafted features are extracted from the middle frame, fused, and regressed via an MLP.

4.3.3. Fusion and Prediction

The 512 deep features and 5 handcrafted features are concatenated and passed through an MLP:

$$\text{Input (517-d)} \rightarrow \text{FC(128)} \rightarrow \text{ReLU} \rightarrow \text{FC(1)}$$

4.3.4. Training Details

The model is trained on the [20] dataset using the Adam optimizer with a learning rate of 1×10^{-4} for 1000 epochs. Batch size is set to 8. A hybrid loss function is used:

$$\mathcal{L}_{\text{hybrid}} = \text{MSE} + \alpha \cdot (1 - \text{SROCC}), \quad \alpha = 0.2$$

4.4. ob1to Team

The flow of the method we used is shown in Fig.5. Firstly, the input screen content images are cropped into fixed-size image patches to facilitate processing of input images with different resolutions. During training, we set the image patch size to 64×64. Secondly, the input image patches undergo multiple stages of multi-scale feature extraction operations, each stage consists of multiple convolutional layers and a max pooling layer.

Then, the dimension of the features extracted in all stages is reduced by adaptive mean pooling and adaptive std pooling to extract global information and feature change information. Next, these feature maps of different scales are concatenated in the channel dimension and fused by convolution to obtain preliminary quality assessment features.

Preliminary quality assessment features should include semantic information and distortion information of the image, so we need to decouple these two types of information. Therefore, we use two convolutional layers with activation functions to decompose the quality assessment features into semantic features and distortion features.

The contribution of distortion features in different dimensions to the final quality score varies, and the semantic information of the image directly affects human perception and evaluation of image quality.

Therefore, we can determine the attention weights of distortion features in various dimensions thru semantic features. Specifically, semantic features dynamically adjust the attention weights of each channel of distortion features thru the attention mechanism (SELayer), thereby enabling the model to better understand the quality assessment issues of images with different content.

Finally, we combine the adjusted distortion features with the semantic features and input them into the MLP for the final regression of image quality scoring.

4.5. 410lab Team

The model is used for screen image quality assessment and prediction by extracting multimodal features from video frames. It takes a set of video frames as input and processes them through three parallel modules: the 3D CNN [3] for modeling temporal dynamic features, and the EasyOCR [22] for extracting textual information,the Swin [33] Transformer for extracting spatial features. The three types of features are stitched together after extraction and passed through two fully connected layers to finally output a quality prediction score.

The network structure of the method is shown in Fig. 6.

5. Teams and Affiliations

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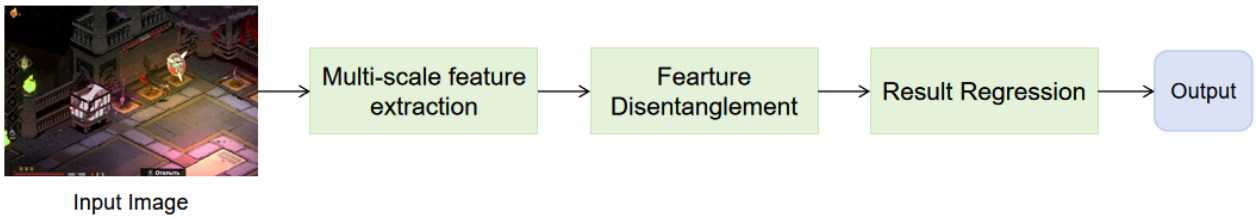


Figure 5. ob1to Team solution.

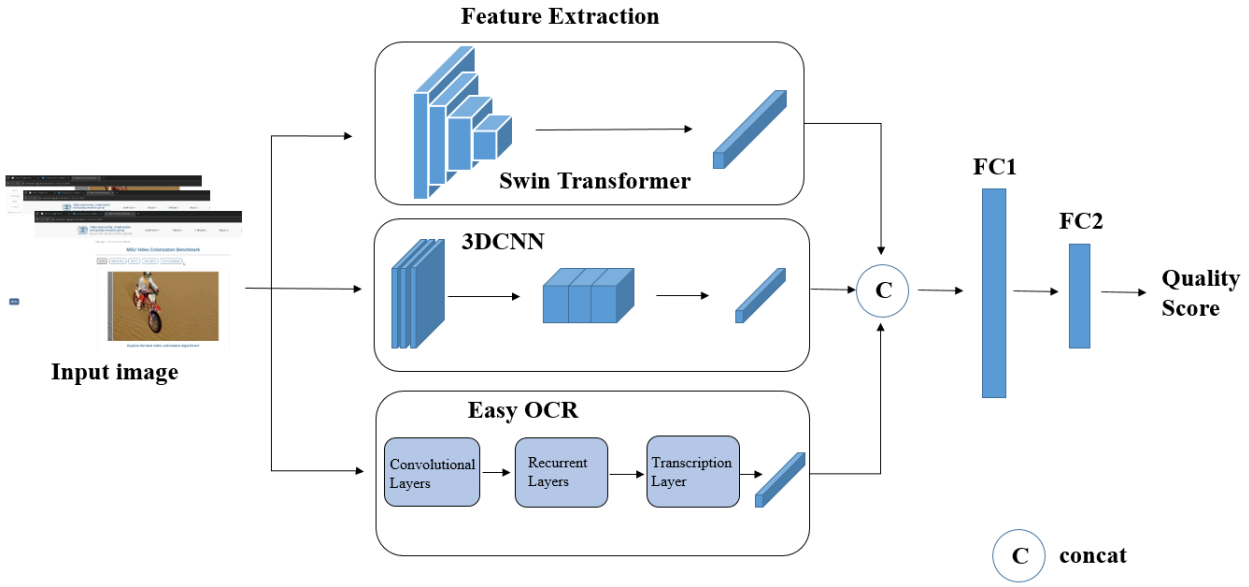


Figure 6. 410lab Team solution: Architecture of the proposed method.

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